Transformer Architecture and the Attention Mechanism

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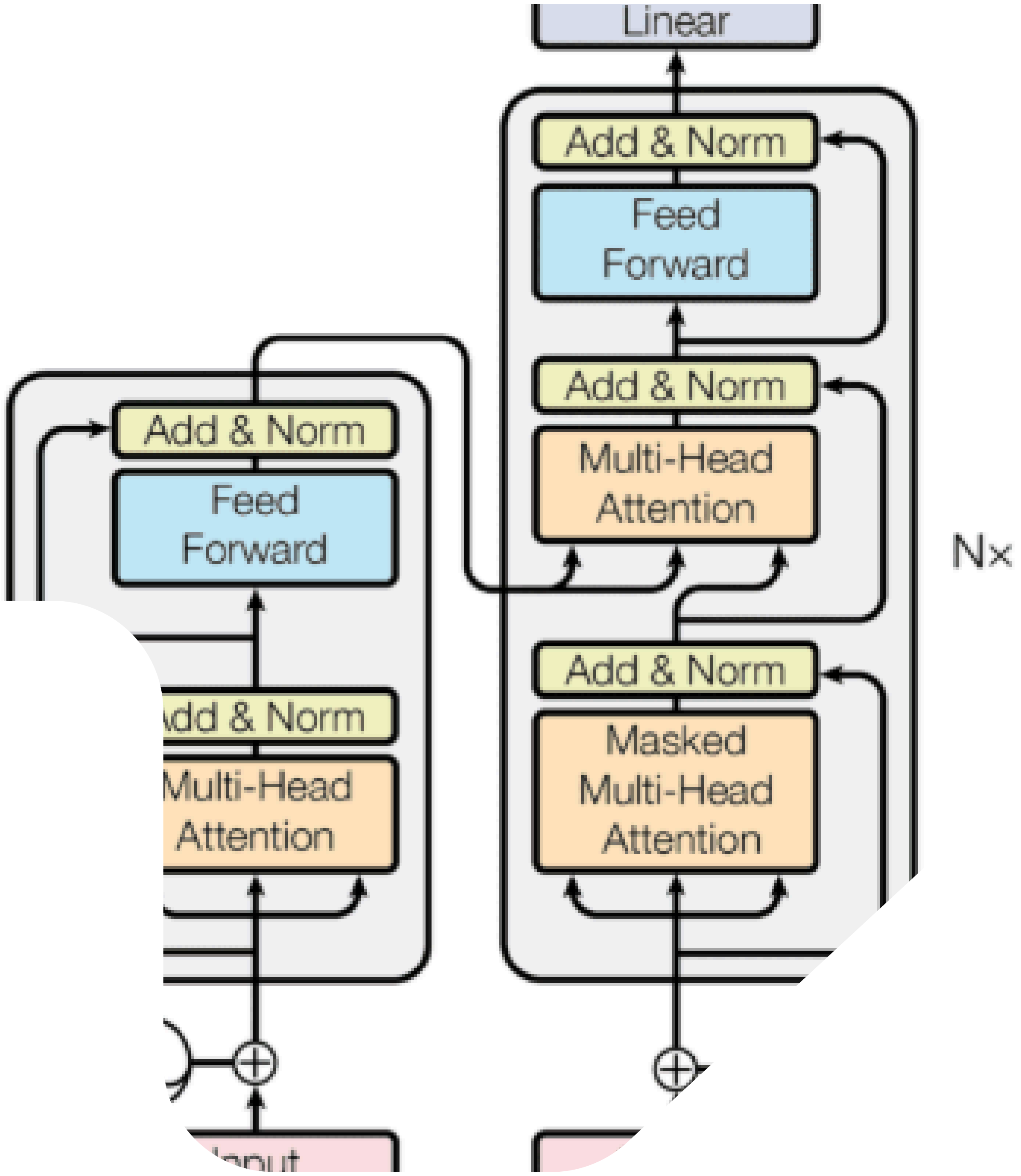
How Artificial Intelligence is Transforming the World.



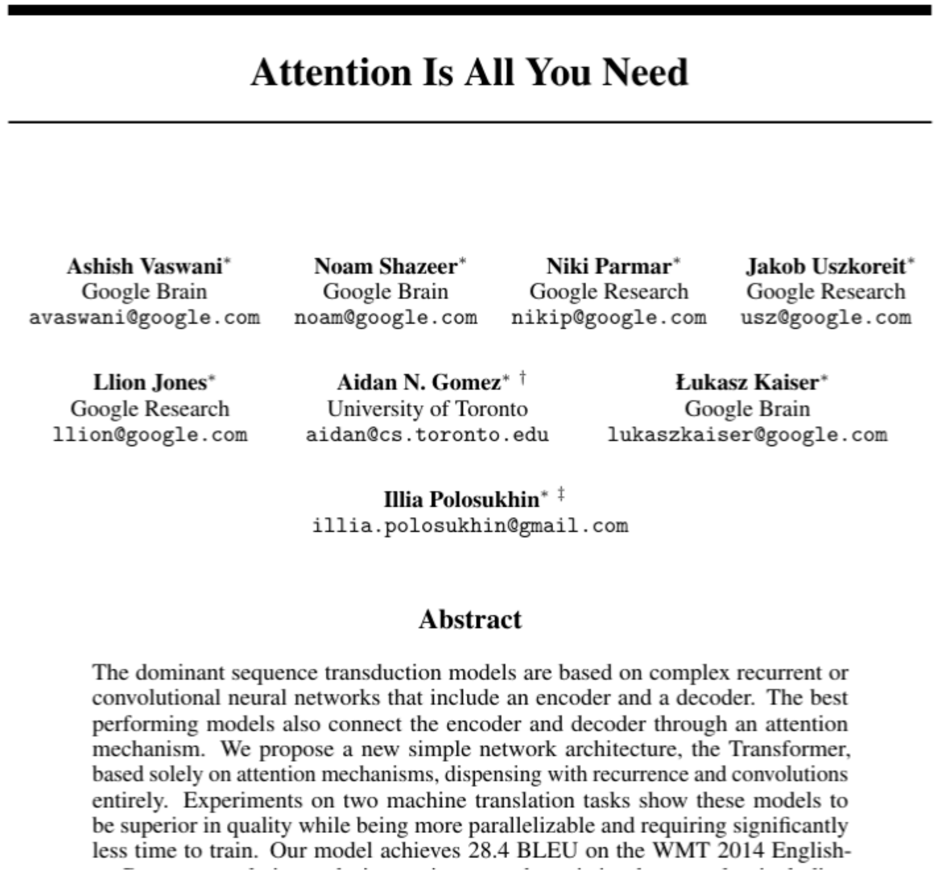
1 The Problem with Previous Models

RNN - Recurrent Neural Networks

- Sequential processing (word by word)
- Hidden state bottleneck

LSTM - Long Short-Term Memory

- Forget gate
- Input gate
- Output gate


Attention Is All You Need

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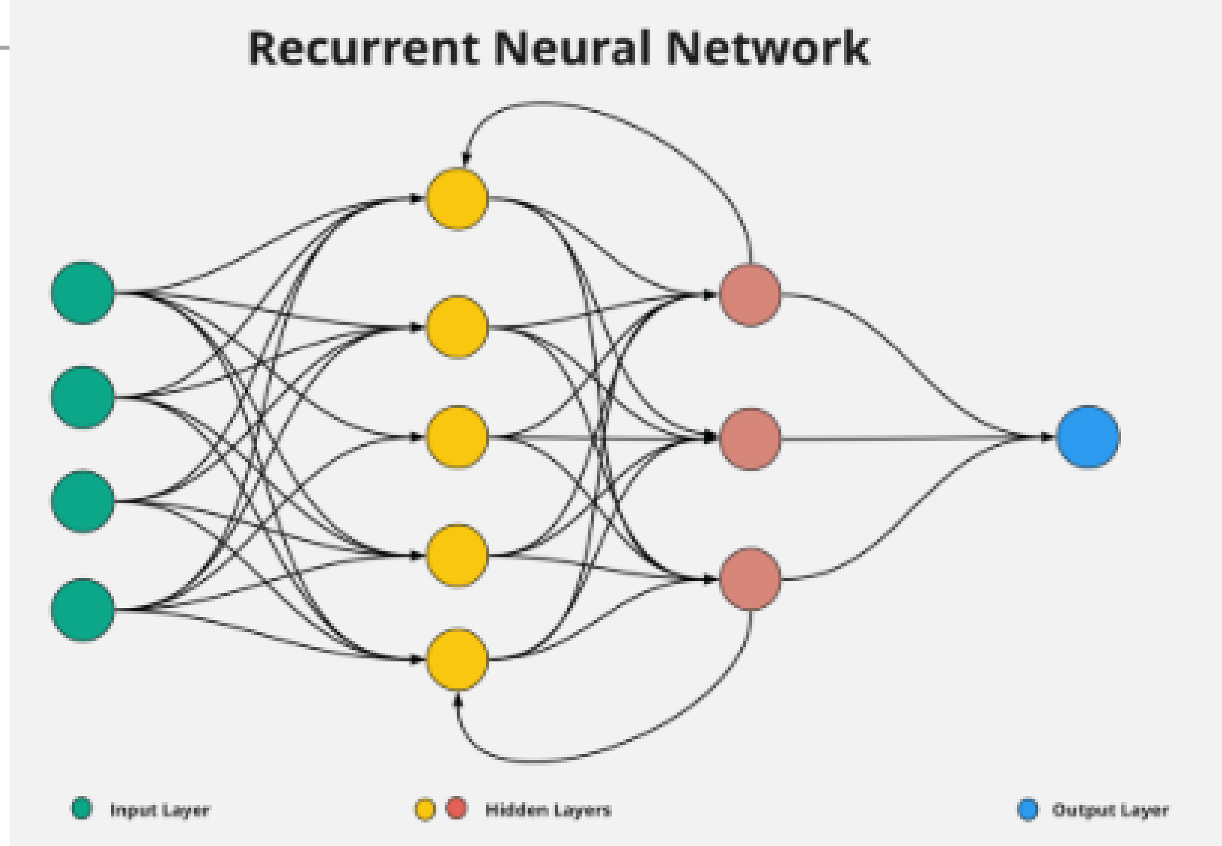
Abstract

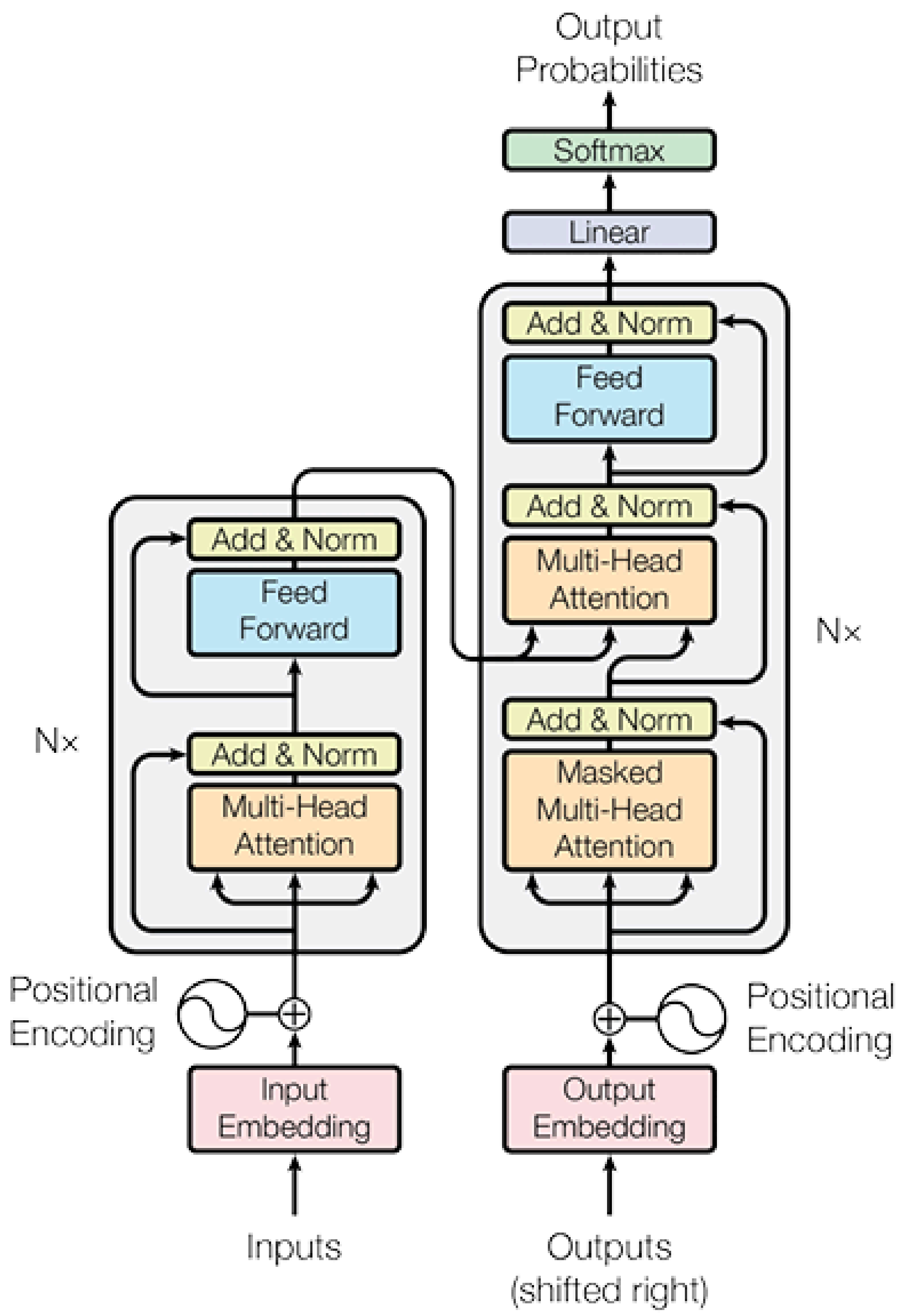
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-



2.Self-Attention

Transformer Architecture

- Parallelization – processing the entire sentence at once
- GPU acceleration
- Equal distance between tokens



3. Transformer Types

Encoder-Decoder (Original)

- Encoder - Input
- Decoder - Output
- Translation and Text Summarization
- BART
- “Understand A, to create B”

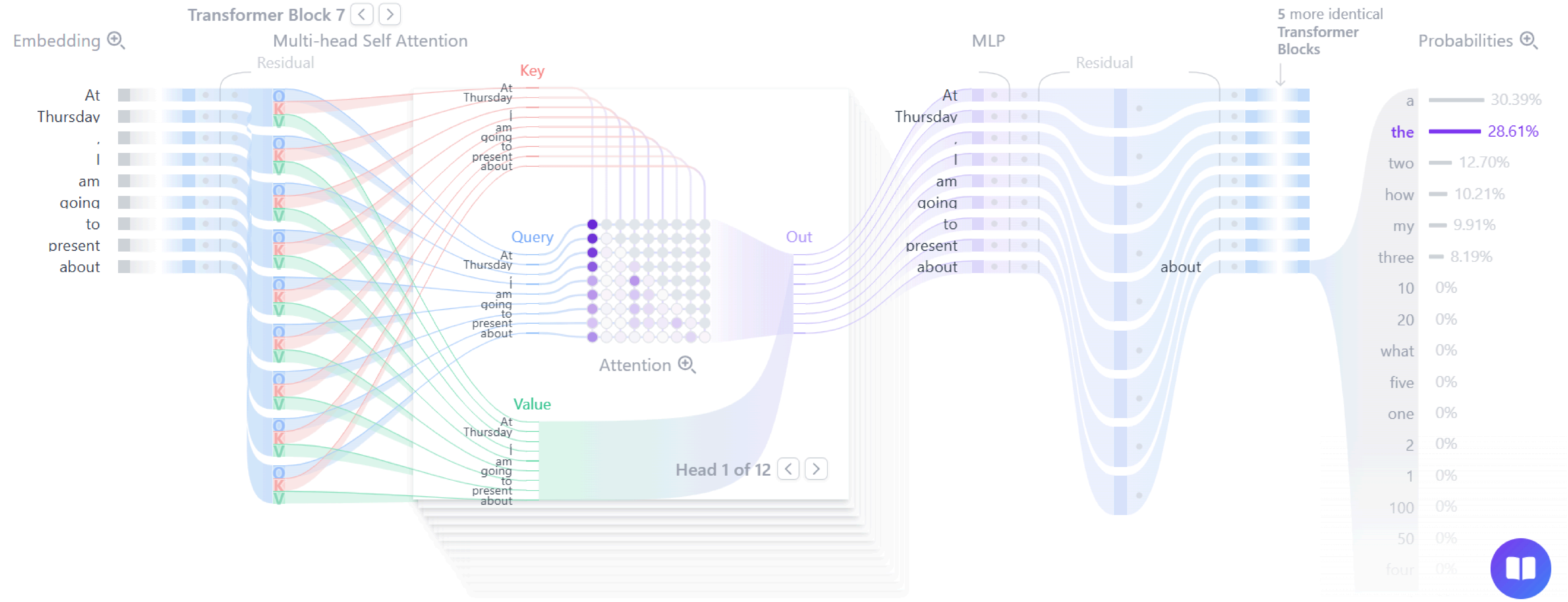
Encoder-only

- Understanding Models
- MLM (Masked Language Modeling)
- Text Classification
- BERT
- “Give me the text, and I will tell you what it means”

Decoder-only

- Generativni modeli
- CLM (Casual Language Modeling)
- Text generation, chatbots, coding...
- GPT
- “Based on what you said, I will continue the sequence”

4. Architecture

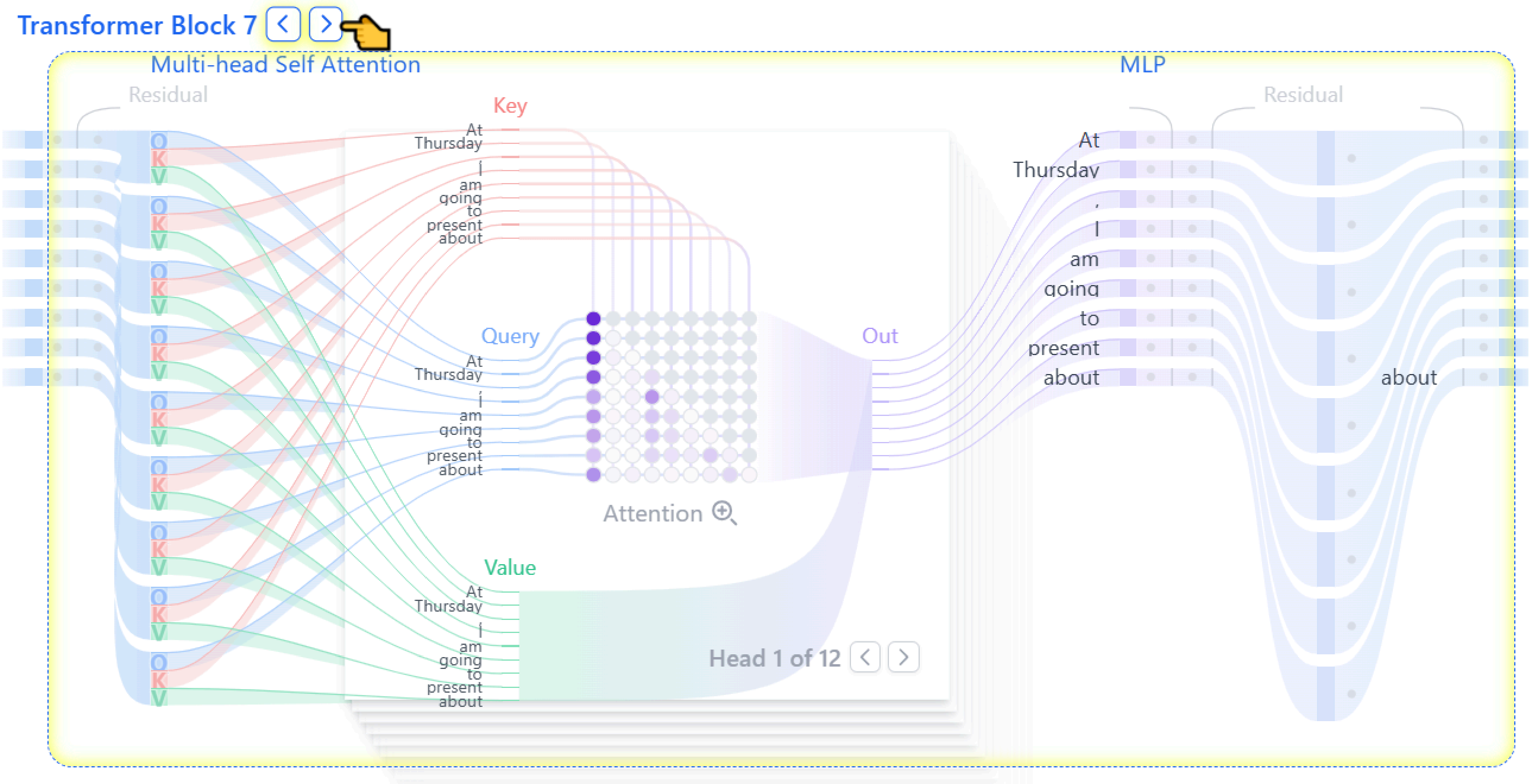


4.1. Embedding

- Tokenization
- Token Embedding
- Positional Encoding
- Final Embeding

4.2. Transformer block

- Heart of the process
- GPT-3 has96 blocks
- Multi-head Self Attention
- MLP

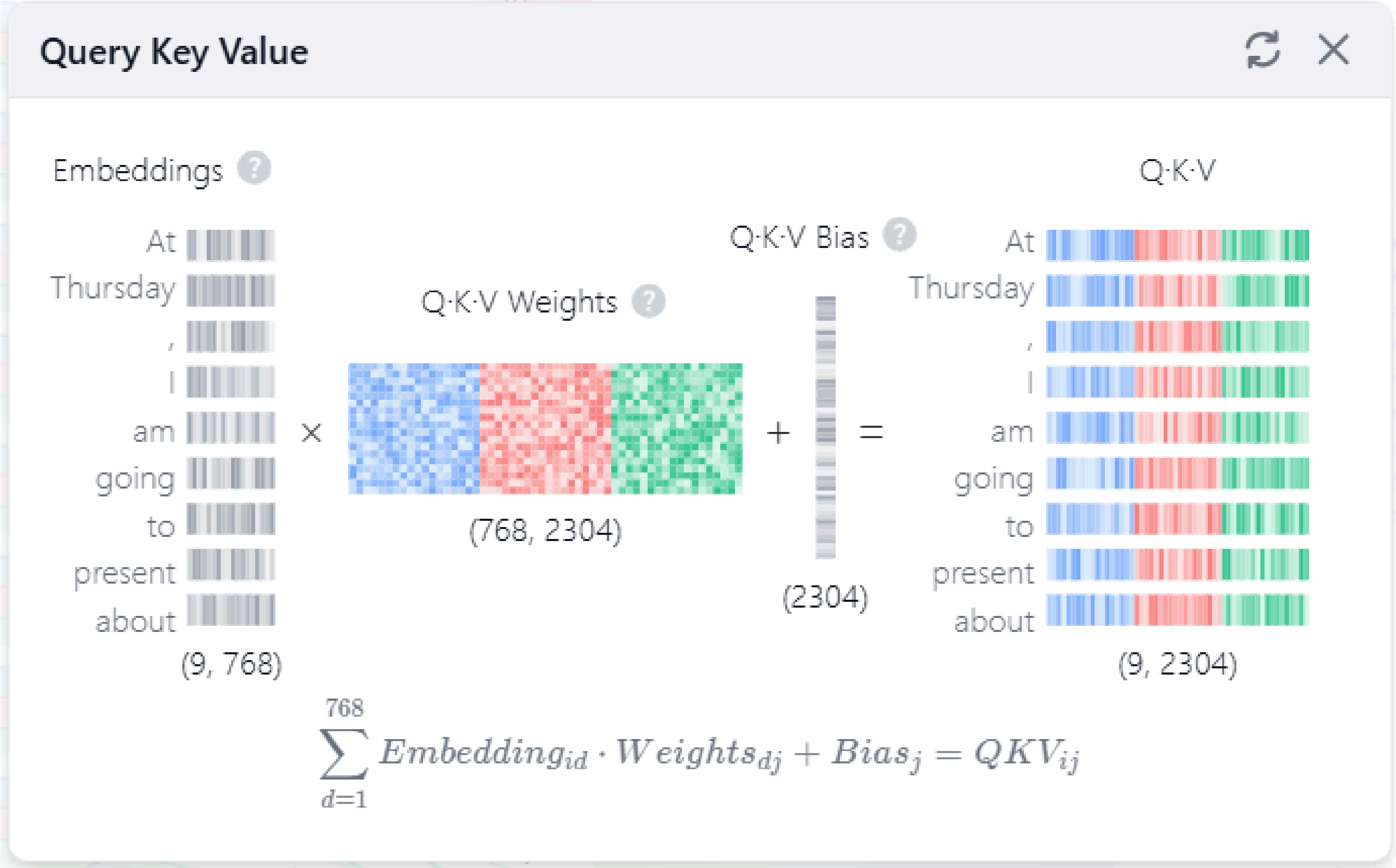


4. 2.1 Multi-head Self-Attention block

Step 1. Q, V, K matrices

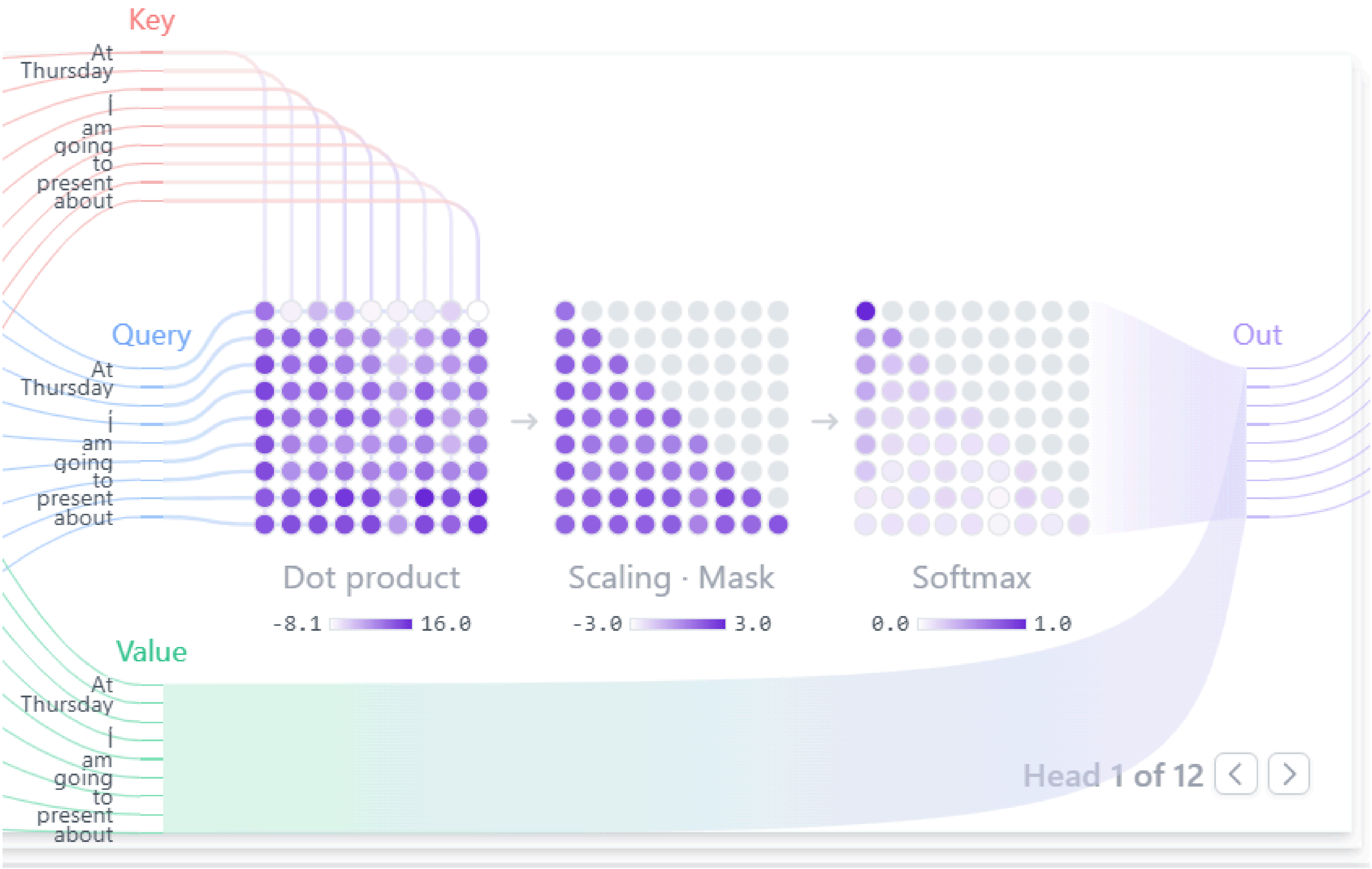
Step 2. Multi-Head splitting

- 2304/12/3 = 64



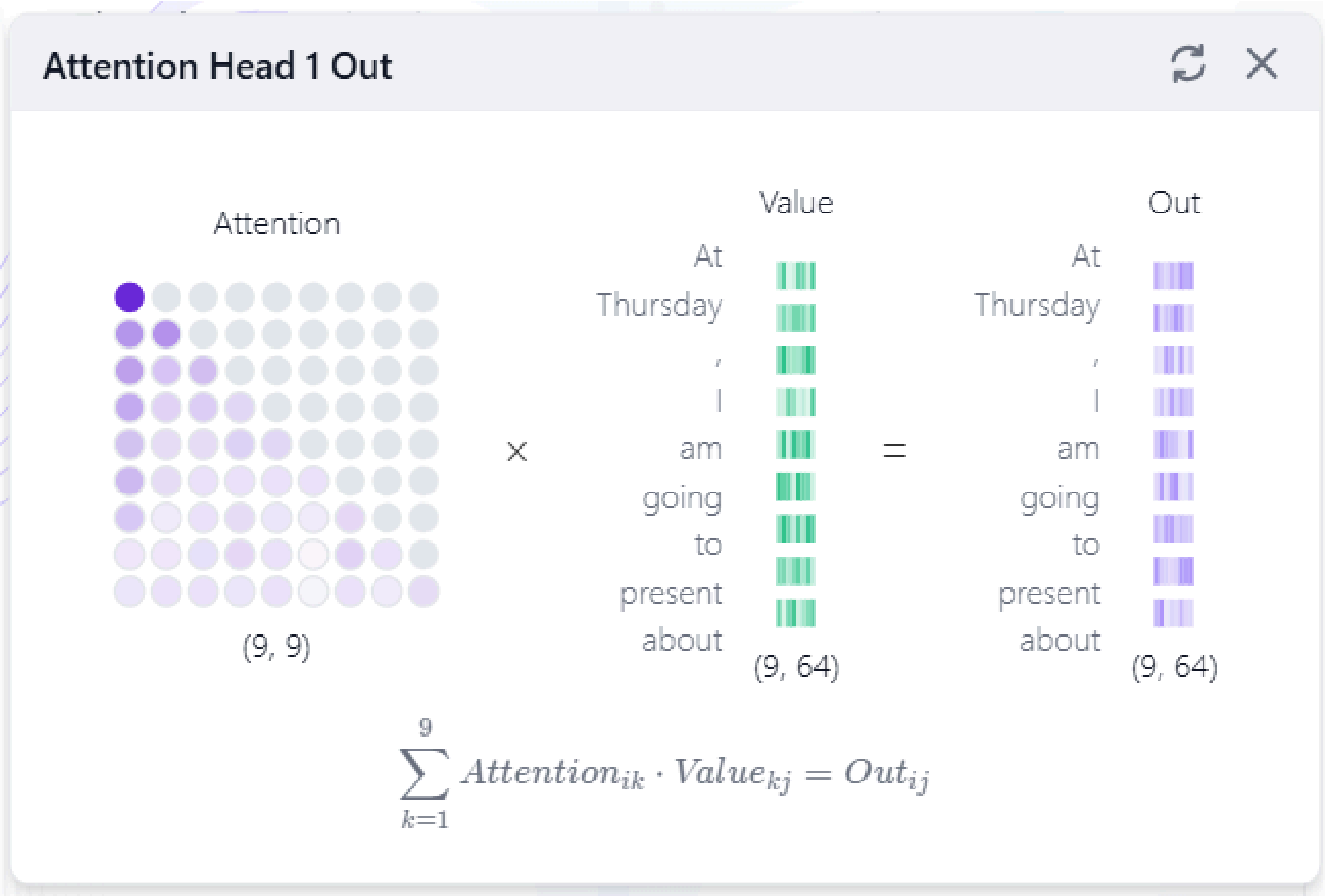
4.2.1 Multi-head Self-Attention block

Step 3. Masked Self Attention



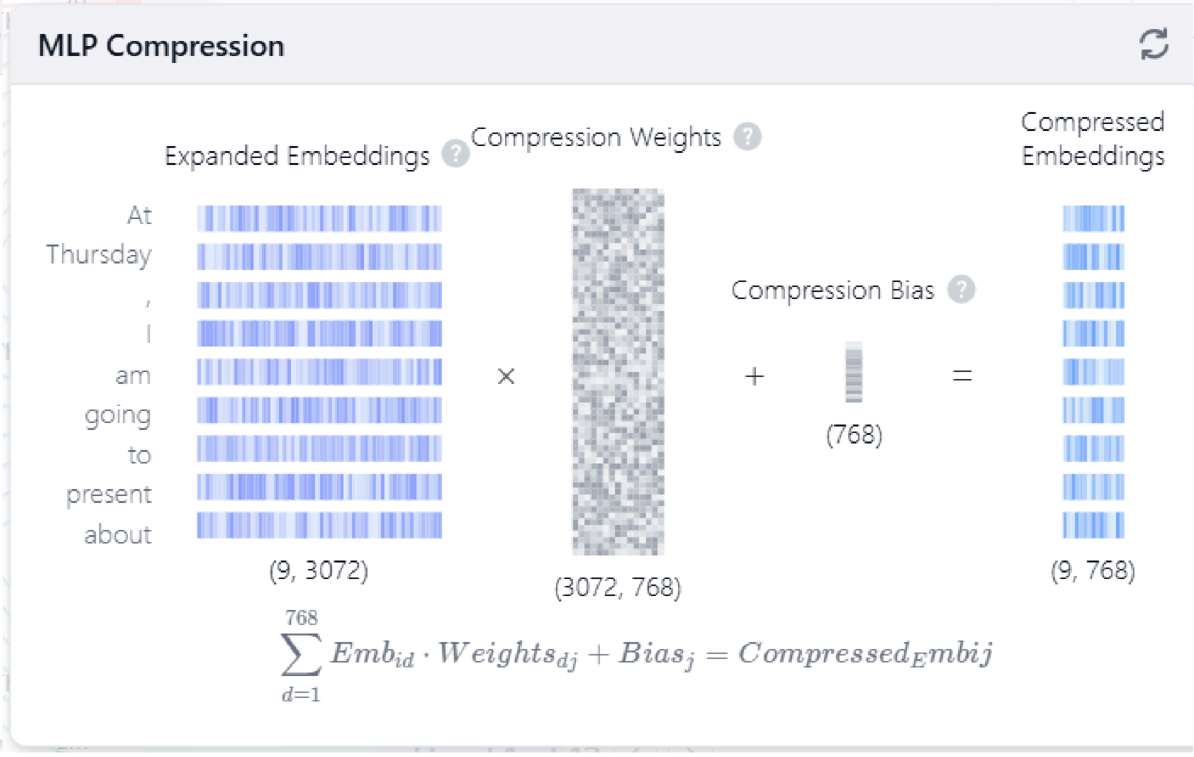
4.2.1 Multi-head Self-Attention block

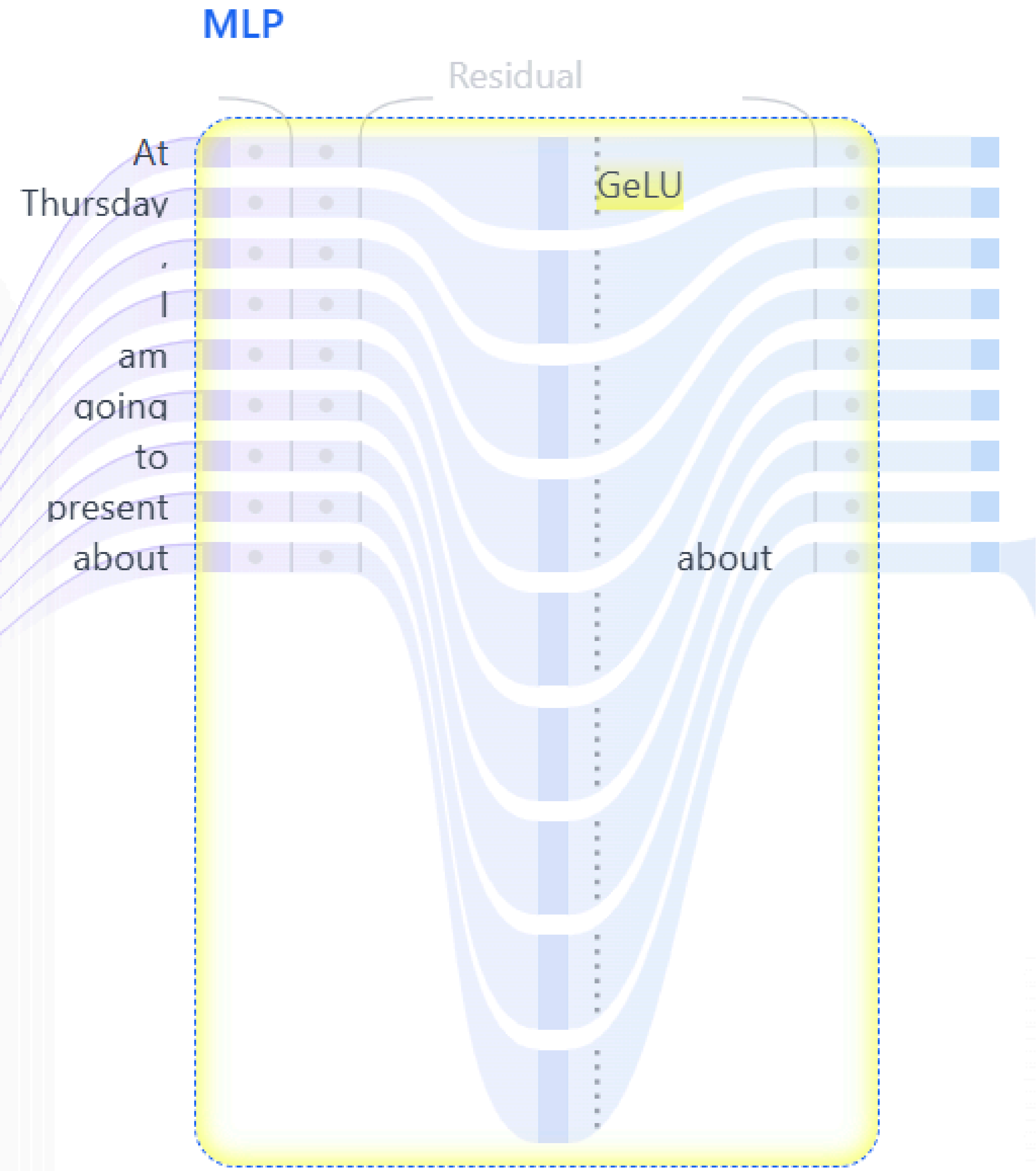
Step 4. Output and head concatenation



4.2.2 MLP

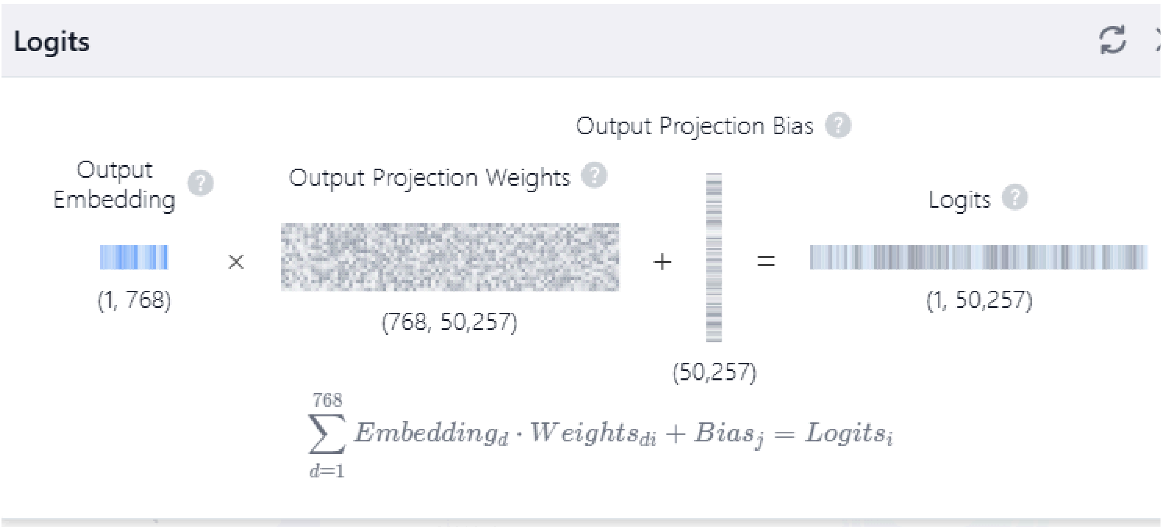
- Normalization and Residual Connection
- Expansion Layer
- GeLU
- Compression Layer
- Normalization and Residual Connection
- Repeat

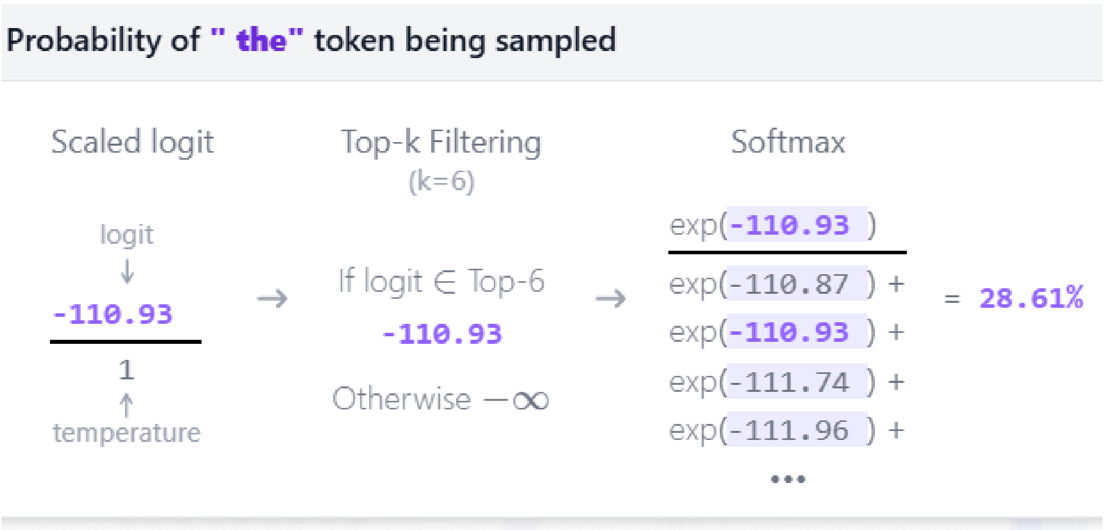


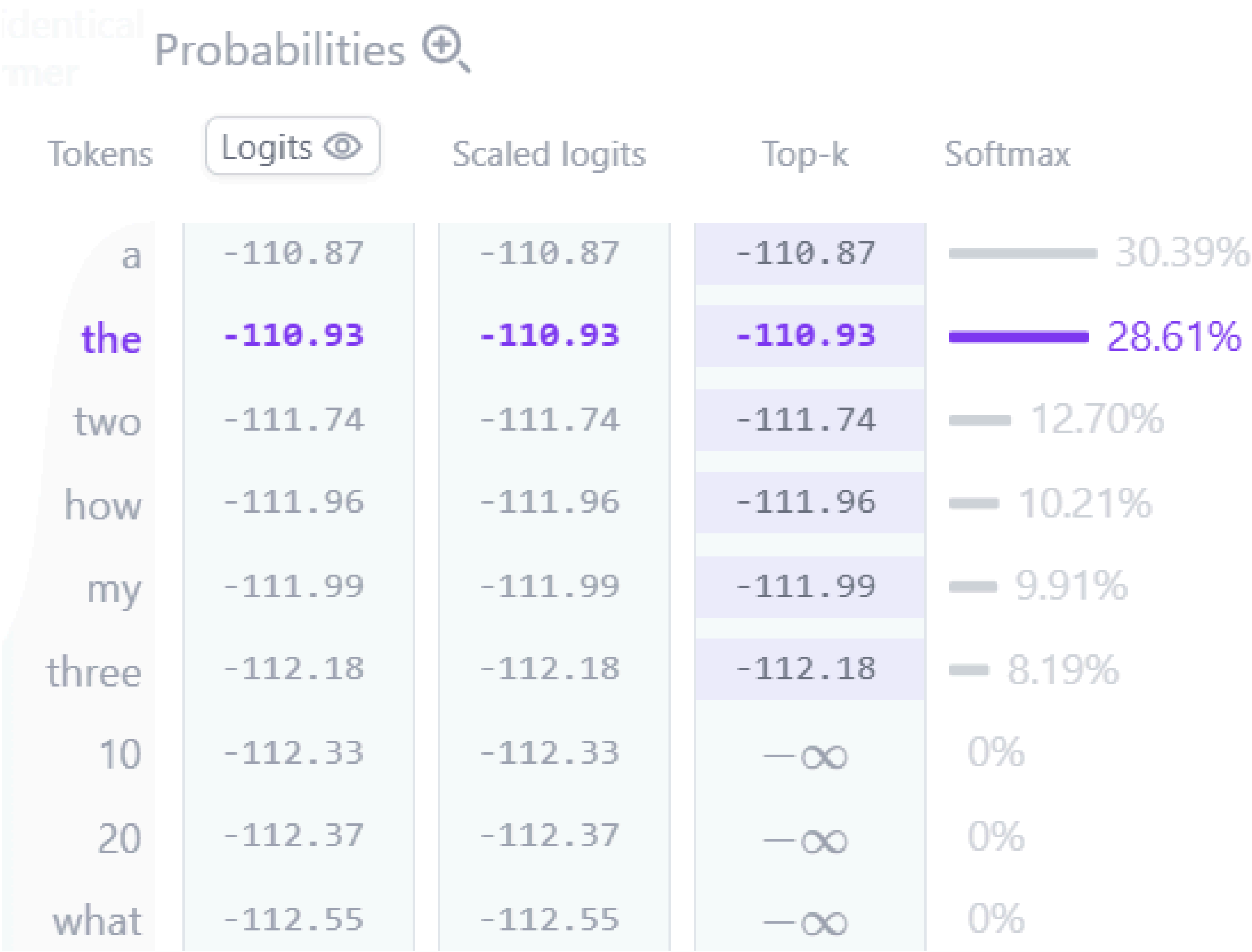


4.2.3 Output

- Final Layer Norm
- Logits
- Softmax
- <|endoftext|>
- Max tokens



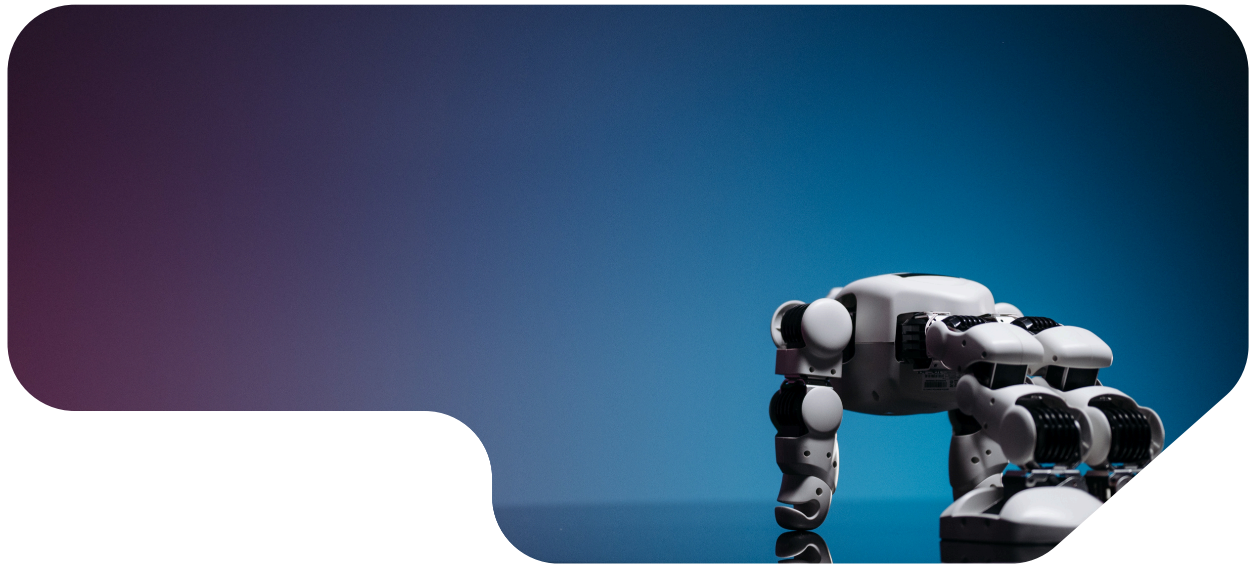




5.Post training

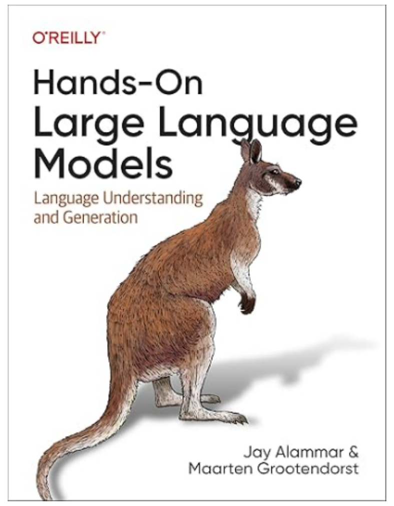
- *SFT (Supervise Fine Tuning)*
- *RLHF (Reinforcement Learning from Human Feedback)*
- *Compression*

6. Conclusion



- *Parallelization*
- *Scalability*
- *Universality*
- *Contextual Understanding*

7. Appendix



- *Hugging Face*
- *Jay Almar*
- *https://poloclub.github.io/transformer-explainer/*
- *"Attention is All You Need"*